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BSCS 6 - C

ARTIFICIAL INTELLIGENCE

MISS UZMA

AI Agent Formulation & Evaluation Assignment

Course Title: Artificial Intelligence

Agent Name: SpamGuard AI (Automated Email Spam Filter)

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1. AI Agent Selection & Name

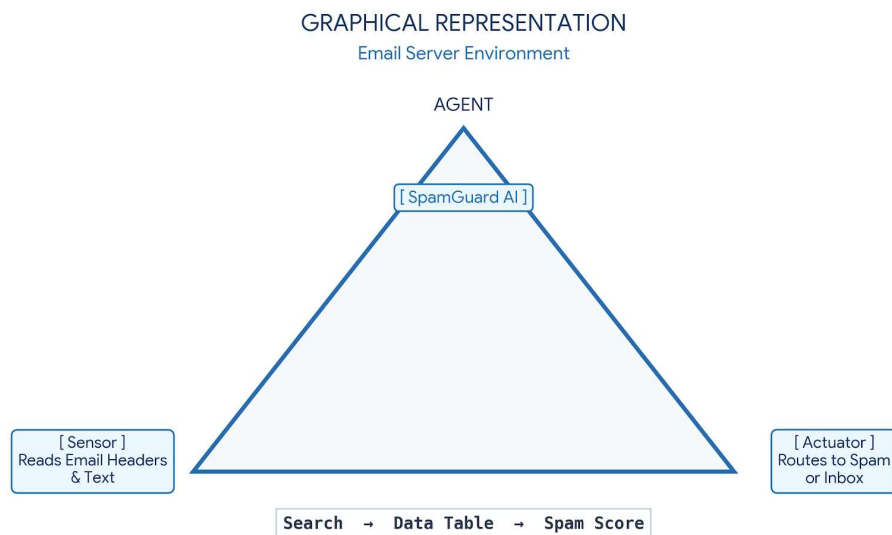
Agent Name: SpamGuard AI

Introduction

An AI agent is a program that reads information from its surroundings using sensors and takes action using actuators to solve a problem. In computer science, an email spam filter is a common example of a software AI agent. SpamGuard AI reads incoming emails, decides if they are safe or spam, and sorts them into the correct inbox folder automatically so the user doesn't have to do it manually.

2. Agent Formulation

• Graphical Representation



• Textation

1. **Sensor:** Collects email text, subject lines, and sender headers from incoming messages.
2. **Actuator:** Executes the routing action to place the email into the Spam Folder or Inbox based on safety rules.
3. **Environment:** Checks the incoming email data stream to determine if an email is safe or spam.

3. PEAS Framework Evaluation

P	Spam Filter Accuracy & Efficient Detection
E	Email Server Data & User Inbox
A	Inbox & Spam Folder Routers
S	Email Text & Header Reader

4. Environment Type

Property	Type
Observability	Partially
Determinism	Stochastic
Episodes	Sequential
Change	Dynamic
Values	Discrete
Agents	Single

Conclusion

SpamGuard AI shows how a simple software agent works in a changing environment. By using the PEAS framework, it catches unwanted spam emails while keeping important emails safe in the main inbox. Designing agents with clear sensors, actuators, and environment types gives us a clear template for building smart computer tools.